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3 HOW SHOULD ENGINEERS THINK ABOUT ETHICS?

EVERY person experiences ethical dilemmas both large and small: How should I treat my dying grandfather? Should I give money to the homeless beggar? Should I use information that I am not supposed to have in preparing my homework or for an exam? In chapter 2 we established that whatever ethical challenges engineers face *as persons*, they also have distinct ethical obligations and commitments *as engineers*. Codes of ethics and professional conduct suggest that engineering ethics is a matter of professional ethics. Although this suggestion is right—that is, engineers should think of engineering ethics as involving their responsibilities as professionals—there is more to engineering ethics than adherence to a professional code of conduct. Not only the scope of engineering ethics but the relationship between engineering ethics and ethics more generally needs further exploration.

In this chapter, we consider several ways of thinking about ethics. First, the notion of professional ethics is expanded beyond codes of conduct to embrace a deeper connection between engineering and society. Engineers have a broad responsibility to consider the social implications of their work. Second, concepts and frameworks from theoretical ethics are introduced as systematic ways of thinking about ethics. Ethical concepts and theories

are relevant broadly, not just to engineering, and they are put forward here as ways of thinking about ethics that are relevant to personal matters as well as to professional decision-making and broader social and policy issues. Last, in this chapter, some attention is given to practical ethics—that is, to how engineers should figure out and decide what to do in real-world situations. Here the suggestion is that ethical decision-making is not just a matter of applying codes of ethics or ethical theories as if they were algorithms but is much more like the activity of designing solutions that fit the context. In real-world situations, one must design courses of action that take a variety of considerations into account, and this often involves making difficult trade-offs.

Although this chapter has an overarching question—how should engineers think about ethics?—it is not framed as a single debate. Professional ethics, theoretical ethics, and practical ethics are not competing perspectives: they are three different ways to think about the ethical challenges that engineers face. Within each perspective there are, however, many debatable issues. For example, in discussing professional ethics, although it is broadly agreed that engineers have some responsibility for the social consequences of their work, there are disputes about what is encompassed in this responsibility and the lengths to which an engineer should be expected to go to fulfill it. As well, in discussing ethical theories, the adequacy of each theory can be debated. And when it comes to practical ethics, the idea that ethical decision-making is a matter of deducing answers from ethical codes or principles or an open-ended process similar to solving design problems can be debated.

AN EXPANDED NOTION OF PROFESSIONAL ETHICS

As suggested, one way to think about engineering ethics is as professional ethics. Using this approach, engineering ethics is focused on how individual engineers and engineers collectively should behave—what they should care about, what they should refrain from doing, and what they must do. Whistleblowing, a classic topic in engineering ethics, exemplifies this. Whistleblowing

typically involves a decision to go against the wishes of a client or an employer; this seems to violate the obligation to act faithfully on the client or employer's behalf in order to fulfill the obligation to protect the public. So, there is nothing wrong with saying that engineering ethics is a matter of professional ethics. What would be wrong is to say that engineering ethics is merely or exclusively a matter of fulfilling one's professional duties as specified in professional codes of conduct. This is much too narrow a conception of engineering ethics insofar as it neglects the deep connection between engineering and people—engineering and society—which is the basis for many of the principles in the codes of ethics.

The connection between engineering and people is not just a connection between engineering and individuals. Engineering affects the lives of individuals, but it also affects the quality and nature of the societies people live in and the distribution of benefits and burdens among individuals and groups. A few decades ago, many engineers rejected the idea that there was any significant connection between engineering and ethics. Claims like the following were commonplace:

"The job of engineers is to figure out how nature works and to manipulate natural phenomena in order to develop useful products and processes."

"The engineer's job is to get the technical right; it is not the engineer's job to worry about society."

"Engineers provide technological solutions, and others decide whether the solutions should be used."

"Technology is neutral; values come in only in the way people use it."

"Since engineering is about machines and devices, it has nothing to do with ethics."

Such claims are rarely made today and, if made, are generally rejected. They are worth mentioning here because they reflect a deep but false presumption that technical and social matters are separate. Today, engineers as well as the public readily recognize that engineering, technology, and society are intimately intertwined. Changes in one can result from and affect the other.

There are at least two important ways in which the intertwining needs to be acknowledged in thinking about engineering ethics. First, engineering is a social activity. It involves people working with one another to accomplish tasks; it involves human beings organized in distinctive ways, for example, in teams and organizational hierarchies. As well, engineering is influenced by the world around it. The history of engineering shows that engineering has changed over the years not just because of technological developments but also because of social ideas and forces that affected the organization of the profession and how engineers work. The emergence of professional societies and the development of licensing procedures are two examples of social organizational change that have affected engineering practice.

Second, engineering and technology powerfully affect the way people live. Engineering and technology affect the quality and character of human living and human experience. They affect what people do and how, where, and when they do it. Bridges and highways affect where people go; cell phones have changed who interacts with whom, when, and how often; biomedical devices affect who lives and dies. Whether one considers global relations enabled by a vast global communication network or our most intimate personal relations affected, for example, by reproductive technologies, in the twenty-first century it is hard to imagine any dimension of human life that has not been influenced by technology and engineering.

So, engineering/technology and society are intimately intertwined. In fact, they are so tightly connected that the phrase “technology affects society” does not adequately capture the connection. For one thing, the influence is bidirectional. Engineering and technology influence the character of society *and* society influences the character of engineering and technology. For another, technology and human behavior work together to make technologies work. Consider large-scale technological endeavors such as nuclear power plants or Facebook. We may refer to these as technological, but their operation depends both on machines and devices *and* on humans behaving in certain ways. In the case of Facebook, for example, you may think it consists primarily of

software, a graphical interface, computers, and monitors, but in order for Facebook to function, many people have to go to work each day to maintain the operation of the system. It is not just those who maintain the computer servers; for Facebook to work, a complex organization of people is essential, including top-level administrators, administrative assistants, people who manage finances, human resource officers, building maintenance workers, and, of course, users.

Once the deep connection between engineering and people/society is acknowledged, it is a small step to ethics. Ethics has to do with how human beings treat one another, and since engineering/technology affect the ways in which people organize themselves and behave toward one another, engineering is an inherently ethical endeavor. Ethical questions can be raised about the nature of the social relationships involved in engineering and the nature of the effects on people that engineering produces. This means that engineering ethics has to do with how engineers conduct themselves in their work *and* with how their behavior and their activities as engineers affect people and social arrangements. It is worth mentioning here that the work of engineers also affects the natural environment and nonhuman living creatures; this, too, is part of engineering ethics.

The Scope of Responsibility for Social Consequences

Although the connection between engineering/technology and people/society is undeniable, a skeptic might challenge the implications of this connection for the social responsibility of individual engineers. That is, a skeptic might claim that individual engineers should not be held responsible for the social implications of their work. In defending this claim, the skeptic might make several points. First, engineers are not the only actors making decisions that affect the character of human societies. In fact, engineers are not even the only actors determining what technologies are developed and how they are used. Many other actors are involved—entrepreneurs, financiers, regulators, marketers, media, environmental groups, and consumers, to name a few.

Second, engineers are generally distant from the social effects of their work. Often they can't see how their work will be used, let alone see the effects it will have. For example, an engineer working on the development of a biomedical device may work primarily in a laboratory and rarely see the people whose lives will be significantly improved if the device is successfully developed. In this case, the connection to people is closer and the effect positive. In contrast, engineers working on such things as transistors or turbines or new materials may have little idea how their work will ultimately get used in a product or process or how it will be put together with other parts or processes. Hence, it is difficult to see how their work will affect what people do.

Third, engineers often work on small parts of large endeavors, so they cannot foresee how the whole to which they contribute will turn out. This means, again, that engineers cannot foresee the social implications of their work. Engineers may be genuinely uncertain as to how their work is going to be used and therefore uncertain as to its effects on individuals or society.

The points that the skeptic makes here are not wrong. In fact, they illustrate the complexity of engineering practice and the challenge of being an ethical engineer. Yes, engineers work with others so that the effects of their efforts are a combination of their effort and that of others. Yes, the effects of engineers' work are often distant, so it takes some effort and even imagination to foresee the consequences. And, yes, engineers often work on only a small component of a larger endeavor, so, again, it is difficult to see what the consequences will be. All of this is true and yet none of it justifies letting engineers off the hook of responsibility. Indeed, it would be dangerous to absolve engineers of responsibility, for to do so is to suggest that engineers should work blindly. It is to suggest that they should be slaves to their bosses or employers—just do what they are told without asking questions—and, perhaps more important, that they should cease to be moral beings when they work. In a sense, such a view frames engineers as machines; they are to work but have no thoughts about what they are doing.

Rather than absolving engineers of responsibility, the points that the skeptics make provide a strong basis for just the opposite

conclusion. Because of the complexity of their relationship to the effects of their work, engineers have a responsibility to make a special effort to keep in mind the connection between their work and the people who will be affected by it. Engineers should think about and inquire about the contexts in which they are working. No matter whom they are working for or on what project they are working, there are always some ideas about how the work will be used, whom it will serve, and what its benefits will be. Generally, projects are pursued precisely because someone has ideas about what the project will accomplish, whom it will help, and how it will help. Engineers generally don't and shouldn't ignore these ideas.

And, of course, many engineers work on projects or do jobs in which they are not distant from the people affected and the social implications. They know who is likely to be affected and how. For example, an engineer may know the neighborhood in which a building is being constructed, the industry in which a mechanical device or process will be used, the types of consumers that are targeted to benefit, and so on. In many cases, engineers not only know who will be affected but also have some idea of the risks and which groups of people are likely to bear those risks.

So, there are compelling reasons not to let engineers off the hook of responsibility for the work that they do. This responsibility can be acted on in a wide range of situations, for example, when engineers decide what job to take or what project to work on or when they decide whether to pursue a suspicion they have about a potential danger or risk or whether to report wrongdoing. Even though the social implications of an individual engineer's work may be distant and difficult to anticipate, the connection between engineering and society is always there, and acknowledging the connection is essential to understanding the social significance of what one is doing. Only when engineers understand this connection can they fully understand engineering ethics.

We will return to this topic in chapter 8 when we explore engineers' responsibilities for social justice. For now, one important caveat is worth noting. The focus in the preceding discussion has been on individual engineers, and the story is quite different when it comes to engineers acting collectively. Collectively engineers

have more power to address the social implications of large engineering endeavors such as transportation systems or energy systems and their safety and their effects on health and the environment. So, the connection between engineering and society has implications for engineers collectively as well as for individual engineers.

THEORETICAL ETHICS

Setting aside the “engineering” part of engineering ethics, there is a long tradition of ethical theory and ethical thought that is relevant to engineers’ thinking about ethics. In its deepest sense, ethics has to do with how human beings should treat one another, right and wrong, good and bad, what we owe one another, and what is just or fair. As such it is an inherently normative domain; it focuses on what should happen, not what does happen. For these normative matters, moral philosophy provides a systematic body of thought. For centuries, moral philosophers (Western and Eastern alike) have sought to understand the foundations—the underpinnings—of morality. In the past several decades a substantial amount of attention in engineering ethics has been focused on three major theories: utilitarianism, Kantian ethics, and virtue ethics.

These three theories are introduced here as examples of theoretical, systematic thinking about ethics. They are not the be all or end all of moral philosophy and should not be taken as such. They are introduced here as starting places for thinking about ethics theoretically. Each of the theories provides a framework, a set of concepts, and language that are useful for discussion and understanding of ethical issues and dilemmas. Each theory illustrates a way in which ethical analysis can bring better understanding.

Utilitarianism

Utilitarianism was developed by philosopher Jeremy Bentham (1748–1832) and elaborated and extended by John Stuart Mill (1806–1873). According to utilitarianism, what makes behavior

right or wrong depends on the consequences. For this reason, utilitarianism is also often referred to as a form of consequentialism. What is important according to consequentialists is the outcome or results of behavior and not the intention of the person acting. Traditional utilitarianism is focused on a particular type of consequence—namely, those that produce happiness or well-being. Actions are good when they produce the most happiness and bad when they produce net unhappiness. This philosophy is called utilitarianism because it holds that actions, rules, or policies are good because of their utility in bringing about good consequences.

The fundamental principle of utilitarianism is this: Everyone ought to act so as to bring about the greatest amount of happiness. Utilitarianism recommends that when you are deciding what to do, you should assess the consequences of your acting in one way or another and take the action that will bring about the most overall good consequences.

Why focus on consequences and neglect intentions? Utilitarians argue that happiness is the only thing that is so valuable that it requires no justification. In utilitarianism a distinction is made between things that are valued because they lead to something else (instrumental goods) and things that are valued in themselves—that is, for their own sake (intrinsic goods). Money is a classic example of something that is instrumentally good. It is valuable, not for its own sake, but rather as a means for acquiring other things. On the other hand, intrinsic goods are not valued because they are a means to something else. They are valuable in themselves. Happiness is an intrinsic good. Friendship is also often used as an example of another intrinsic good. We don't value friends because of what they can do for us; we value them in themselves.

According to utilitarians, a moral theory must be based on something that is so valuable that people should commit their lives to it. Happiness, they argue, is the only good that can serve in this role because happiness is the only thing that is truly valued for its own sake. Indeed, some utilitarians claim that all other goods are desired as means to happiness. Even friendship, health,

and knowledge, they would say, are valued because they lead to happiness.

With happiness as the ultimate good, then right and wrong follow; the right action is the one that brings about the most net happiness, and the wrong action is the opposite. When a person faces a decision, the person should consider possible courses of action, anticipate the consequences of each alternative, and choose that action which brings about the most happiness—that is, the most overall net happiness. To be sure, the right action may be one that brings about some negative consequences (unhappiness), but that action is justified if it has the most net happiness or the least net unhappiness of all the alternatives.

Utilitarianism should not be confused with egoism. Egoism is a theory that specifies that one should act so as to bring about the greatest happiness or good consequences *for one's self*. What is good is what makes me happy or gets me what I want. Utilitarianism does not say that one should maximize one's own good. The sum total of happiness in the world is what is at issue. Thus, when you evaluate your alternatives, you have to ask about their effects on the happiness of everyone. This includes effects on you, but your happiness counts the same as the happiness of others. It may turn out to be right for you to do something that will diminish your own happiness because it will bring about a marked increase in overall happiness.

As a general theory, utilitarianism has been influential in such fields as economics and policy analysis. Legislators and public policy makers seek policies that will produce good consequences, and they often opt for policies that may have some negative consequences, but they do so with the idea that on balance there will be more good consequences than bad. Cost-benefit and risk-benefit analysis are rooted in utilitarian thinking insofar as they specify that one should balance risks and benefits against one another in making a decision.

There are many forms of utilitarianism, and one contentious issue is whether the focus should be on *acts* or *rules*. Some utilitarians have recognized that it would be counter to overall happiness if each of us had to calculate what the consequences of

every one of our actions would be before we act. Not only is this impractical—because it is time consuming and because sometimes we must act quickly—but often the consequences are impossible to foresee. There is thus a need for general rules to guide our actions in ordinary situations. Accordingly, some argue that we should focus on rules rather than acts.

Rule utilitarians argue that we should adopt rules that, if followed by everyone, would, in the long run, maximize happiness. Take, for example, telling the truth. If individuals regularly told lies, it would be very disruptive. You would never know when to believe what you were told. In the long run, a rule obligating people to tell the truth has enormous beneficial consequences. Thus, even though there are sometimes occasions when more good will come from lying, the rule “Tell the truth” is justified on utilitarianism. “Keep your promises” and “Don’t reward behavior that causes pain to others” are also rules that can be justified on utilitarian grounds. According to rule utilitarianism, if it is true that when people follow the rule, long-term net good consequences result, then the rule is justified and individuals ought to follow the rule.

Act utilitarians believe that each action should be considered on its own. They believe that even though it may be difficult for us to anticipate the consequences of our actions, that is what we should try to do. Rules can be used, but they are not strict; they should be understood to be rules of thumb—that is, general guidelines that can be abandoned in situations where it is clear that more good consequences will result from breaking them. Rule utilitarians, on the other hand, take rules to be strict. People should follow the rules even if in one particular case, the negative consequences outweigh the benefits. Rule utilitarians worry that breaking the rules on occasion will undermine the effect of rules and will therefore have overall net negative consequences.

Utilitarianism can be used for engineering ethics in many ways. An engineer might use the theory to think through a situation where the right thing to do is not so obvious. The engineer would identify the positive and negative effects of one course of action or another and calculate which would be better. The theory might also be used to understand the justification for rules or

practices such as those specified in the codes of ethics. A rule—such as the one specifying that engineers should avoid deceptive acts—is justified on grounds that overall good consequences result from engineers following it. And good consequences here does not mean consequences only for engineers, it means consequences for engineering and engineers as well as clients, employers, and the public. Another related way that the theory might be used is to make the case for eliminating or changing a professional rule or practice. Suppose, for example, that policy makers want to change an existing practice by requiring the signature of a licensed engineer for an activity that previously could be done without it. A change of this kind might be justified in a utilitarian framework—that is, on grounds that it would reduce the risk of harm, thereby improving the positive effects of the activity.

Critique of Utilitarianism

Utilitarianism is an important framework for thinking about ethics, but it is not without problems. Some have suggested that it does not adequately capture what is important about ethical thinking. Perhaps the most significant criticism is that the theory goes against strongly held moral intuitions insofar as it seems to tolerate, if not recommend, that burdens be imposed on some individuals for the sake of overall good. Although it is true that every person is to be counted equally—that is, no one person's happiness or unhappiness is to be counted more than another's—there are situations in which overall happiness would result from sacrificing the happiness of a few. Suppose, for example, that having a small number of slaves would create great happiness for a large number of individuals. The individuals who were made slaves would be unhappy, but this would be counterbalanced by significant increases in the happiness of many others. Such a situation would seem to be justifiable in a utilitarian framework. Another more contemporary example is to imagine a situation in which by killing one person and using their organs for transplantation, ten lives could be saved. Killing one to save ten would seem to maximize good consequences.

Critics of utilitarianism argue, then, that since utilitarianism justifies practices such as slavery or more broadly sacrificing some for the sake of others, it is not a cogent ethical theory. In their defense, utilitarians point to the difference between short-term and long-term consequences. Utilitarianism is concerned with all consequences, and when long-term consequences are taken into account, such practices as slavery and killing innocent people to save other lives cannot be justified. In the long run, such practices have the effect of creating so much fear in people that net happiness is diminished rather than increased. Imagine the fear and anxiety that would prevail in a society in which anyone might at any time be taken as a slave. Or imagine the reluctance of anyone to go to a hospital if there was even a remote possibility that they might be killed if they happen to be at the hospital at a time when a major accident occurred and organs were needed to save many victims. This defense shows that utilitarianism is not a simple theory—that is, calculating the consequences of particular actions and practices is not as easy as it might at first appear.

Nevertheless, and despite the utilitarian defense in terms of long-term consequences, the concern of critics is that, at its heart, utilitarianism frames individuals as valuable in terms of their social value—that is, the value of a person has to do with the person's contribution to the overall balance of happiness—and that is problematic. By contrast, a strongly held moral principle is that people are valuable in themselves.

This idea of individuals being valuable in themselves is the central idea in deontological or Kantian theory. However, before we take up this alternative theory, we should note that utilitarianism goes a long way in providing a reasoned, systematic, and comprehensive account of morality. It is not a theory to be dismissed lightly. The idea of basing moral decisions on something that is intrinsically valuable is compelling; in some sense, all human beings seem to seek something like happiness or well-being; and consequences are an important element in moral reasoning and in moral practices.

Kantian Theory

In utilitarianism, what makes an action right or wrong is outside the action: it is the consequences of the action. What the person is thinking at the time of the action is irrelevant—that is, little importance is given to what a person intends or seeks to achieve by an action. By contrast, Immanuel Kant (1724–1804) put forward an ethical theory that is focused on the internal character of actions. Because his theory emphasizes acting from duty, it is characterized as a deontological theory. (The word *deontology* derives from the Greek words *deon*, duty, and *logos*, science.) According to Kant, actions that are done from a sense of duty are morally worthy, and those done for self-interest or simply out of habit are not. Thus, if I tell the truth because I fear getting caught in a lie or because I believe that people will think better of me if I do, then my act is not morally valuable. By contrast, if I am in a difficult situation in which telling the truth may get me into trouble and I tell the truth because I recognize that I have a duty not to treat the person asking the question as a means to my well-being, then my action is morally worthy.

Kant gives a thorough account of what it means to act from duty, but before explaining that, it will be helpful to return to utilitarianism. Utilitarianism goes wrong, according to deontologists, when it focuses on happiness as the ultimate good for human beings. Happiness cannot be the highest good for human beings because it is not a unique or special capacity of humans. Humans have the capacity to reason. We are rational beings. This is not to say that humans always do the most rational thing. Rather, the point is that human beings think about how they will act and then choose how they will behave. Many animals and all inanimate objects are entirely controlled by the laws of nature. Plants turn toward the sun because of photosynthesis. Objects fall because of gravity. Water boils when it reaches 212° F (100° C). In contrast, humans have ideas and can make rules and follow them. I can decide to (or not to) exercise every day, lie when asked a question, refuse charity to a beggar, keep my promises, or pay my debts on time. According to Kant, this is the difference between acting

according to law and acting according to the conception of law. Humans are unique insofar as we can act according to the conception of law—that is, according to rules and laws that we recognize.

Importantly, if humans lacked the capacity to reason about their behavior before acting, morality would not be possible. Morality is a set of rules and values that humans adopt to direct their behavior. We can act on our beliefs about how we should behave. Morality can't apply to entities that act by instinct or whose behavior is entirely determined by the laws of nature.

Of course, questions arise about how we should think about infants, those with mental disorders, and animals with varying degrees of intelligence, but the important point is that for Kant, the capacity to give laws to one's self is the foundational principle of morality. Moral behavior both derives from the rational capacity of humans and directs it. Morality would not be possible were it not for the fact that humans have the capacity to choose how to act. If our behavior was entirely controlled (determined) by nature, then we would not be capable of following moral edicts.

For deontologists, the fact that human beings have this capacity means that we should recognize this capacity in every human being—that is, we should treat human beings as beings with the capacity to control their own behavior. This idea is captured in what Kant calls the *categorical imperative*. There are several versions of the imperative, but the first expresses the most fundamental idea: We should never treat another human being merely as means; we should always treat others as ends in themselves. When you treat someone merely as a means to your own ends, you deny their humanity; you deny their capacity to reason for themselves and make choices for themselves. This is precisely what is wrong with slavery and other utilitarian practices that treat some human beings as a means to the happiness of other human beings. Such practices treat human beings as things, not recognizing their humanity.

It is important to note that the categorical imperative does not say that you should never treat another person as a means. It specifies only that you never treat another *merely* as a means. You can use another person as a means to one of your ends as long as

you do so in a way that recognizes the person's humanity. This means many things, but most important, it means respecting each person's autonomy. To illustrate what this means, consider two of the most significant ways in which you can violate the categorical imperative. One is by lying, and the other is by coercing. Suppose that as an engineer you agree to do work for a client. You might, in a sense, be using the client as a means to your end of making money. However, you wouldn't be treating the client merely as a means as long as you didn't lie to or coerce the client. If you lied to the client about, say, your credentials or the safety of the work that you did, then you would not be respecting the client as a being with her own ends. The client has the capacity to decide whether to hire you or to accept your work, and by lying to her, you have manipulated her into serving your ends and have undermined her capacity to serve her ends. Coercion has a similar problem. Engineers rarely coerce their clients, but extortion comes close. Extortion is an example of pressuring a client to do something that he wouldn't choose to do unless threatened. Say, you threatened to harm a client or one of his loved ones unless he gave you a contract to do work. Again, you would be using the client for your ends and not treating the client as a being with his own ends, able to make decisions for himself.

Another way that Kant proposed that we think about the categorical imperative involves universalizability. Kant argued that, implicitly, actions have maxims. A maxim is the principle underlying an action. For example, suppose you promise a contractor to pay her on completion of work on a building contract and then you decide not to pay her right away (say, you discover you had less money in your bank account and would fall below the minimum balance for earning interest if you paid her right away). Were you to try to universalize your action, it would seem that the underlying principle is that it is okay to break a promise when it is inconvenient or somewhat costly to keep it. According to Kant, you should only do those actions the maxim of which you could universalize. You would be unlikely to want to universalize this principle because that would mean that it was okay for everyone to renege on their promises when it was inconvenient to keep

them. Among other things, this would mean that it is okay for those who promised *you* something to renege on their promises. However, Kant's point is not that it is against your interest to universalize this maxim. Rather, universalizing this maxim would, according to Kant, undermine the very meaning of promising. If breaking promises were routine and commonplace, uttering the words "I promise" would be meaningless. And if you can't universalize the maxim of an action, then the action is wrong. Using the categorical imperative in this way, Kantian theory asserts that some actions are always wrong, including promise breaking, lying, and killing—because the maxims of such actions cannot be universalized.

Like utilitarianism, Kantian theory has its critics. As might be anticipated, one line of criticism is to insist that the consequences of actions are important and should not be ignored. Rather than go further into the debate between utilitarians and Kantian theorists, we will turn to another theory that faults both utilitarianism and Kantian ethics for their blindness to the role of virtues in understanding morality. Before we do that, however, a note about a contemporary technological manifestation of the debate between utilitarian and Kantian theory is worthy of mention.

Utilitarianism and Kantian Theory in Autonomous Cars

The difference between utilitarianism and Kantian theory has recently come into play in a controversial engineering problem. In the public discussion and debate about autonomous cars, some attention has been given to how the cars might be programmed to respond in life-threatening situations. Suppose that two autonomous cars are about to crash. The crash is unavoidable, but there is time for some maneuvering in how the cars collide. Sensors make it possible to calculate how many people are in each car and whether any are children. The software of such cars might be designed, following a utilitarian principle, to communicate with the other car or cars involved and have each swerve to minimize the total harm done to all those involved in the accident. On the other hand, some have argued that the software in each

car should be designed to minimize the harm done to passengers in that car. In other words, in deciding how to swerve, the software would only consider the car's passengers and not give priority to any of the persons involved. This seems Kantian in that it treats the passengers in each car as valuable in their own right—that is, as ends in themselves. In support of the Kantian approach, some argue that consumers of autonomous cars wouldn't buy cars that would sacrifice them for overall net good.

This issue may well have to be resolved as autonomous cars become more sophisticated. This will be intriguing to follow since neither ethical theory is simple and, as suggested above, each has ways of dealing with criticisms. Chapter 7 is focused on safety and autonomous cars as an emerging technology.

Virtue Ethics

Both utilitarianism and Kantian ethics are focused on actions: utilitarianism on the consequences resulting from an action, and Kantian ethics on the internal form and motivation of the action. By contrast, virtue ethics focuses on traits of character—the virtues.

According to virtue theory, in thinking about ethics, we should focus on the characteristics that make for a good person. Although there are many virtue traditions, virtue ethics in the West originates with Aristotle (384–322 BCE) in his treatise *Nicomachean Ethics*. Aristotle is a teleological thinker. He argues that in order to find out what is good, we should start by asking about the end or purpose or function (the *telos*) of the entity or activity at issue, which may involve a distinctive function (*ergon*). We can then identify the characteristics that make the entity or activity good for its end. What makes something good is that it has characteristics that are suited to achieve its purpose. For example, if you want to know what constitutes a good automobile, you start by asking what the purpose of a car is. We generally think that a car is supposed to transport a person conveniently and efficiently from one place or another. We also acknowledge that some people select cars in order to make a statement about their personal

identity. So, a good car will be one that is easy to drive, safe, and efficient and makes an appropriate personal statement. These are the excellences of automobiles; automobiles that have these characteristics are best for what cars are supposed to do.

When it comes to ethics, virtue ethicists propose that we use the same reasoning. Ethics has to do with people, so our attention should be focused on what it means to be a good person. To understand this, we should ask: What is the end or essential function of being a human being? And what are the traits that make for living a good life—that is, for flourishing as the essentially social animals that we are? For virtue ethicists, the proper goal of human life, as specified by Aristotle, is *eudaimonia*. This word is typically translated as happiness or well-being or human flourishing. Flourishing is not merely subjective happiness or good feeling; it is an objective and social condition of being and acting well with others. This end is realized through *arete* (the virtues). The virtues are those traits of character that are essential to human well-being, but they are not merely instrumental to human flourishing: they constitute it. A person who reliably exhibits virtues will have a good life.

The virtues are dispositions to behave in a particular way, as appropriate for a particular situation or context. For example, honesty is a virtue, and a person who is honest is one who has the disposition to tell the truth in ways that are appropriate to each situation. Virtues have an enduring quality—one who is honest doesn't just tell the truth on occasion or only when it is convenient. An honest person is one you can count on to tell the truth when the situation calls for it, and to do so in the right way. For example, a person who spreads gossip or inappropriately betrays others' secrets would not be virtuous, even if they were technically telling the truth. An honest person is consistently honest because honesty is part of the person's makeup. The virtues can be thought of as the excellences of a human being. They are the qualities that constitute excellence in a human being. A virtuous person will flourish as a human being. A life in which one behaves with honesty, courage, and generosity, for example, will be a good life, a life constituted by *eudaimonia*.

Virtues are not, of course, inborn; they are developed through practice and habit, and later refined with the aid of one's practical reasoning (*phronesis*). Virtues must be cultivated. They can be developed to some extent as one is growing up, but we all have to work on ourselves to become virtuous. The challenge of being a good person is that of cultivating in yourself the dispositions essential for virtuous action. As one develops, strengthens, and refines these dispositions, one achieves *eudaimonia*.

In recent years, practical ethicists have used virtue theory to think about virtues not just in general but as they may apply to acting in particular roles, such as being a manager or a parent or an engineer. The idea is that when one finds one's self in a situation requiring moral judgment or decision-making, rather than simply looking for a rule or principle or decision procedure to follow, one should consider the virtues associated with being a good X, where X is a particular role.

Although this way of thinking may not always dictate the specific course of action to be taken in a situation, the virtuous person's practical reasoning will help direct the person to an appropriate form of behavior. For example, if an engineer is in a situation in which a client has asked her to do something that she thinks is unsafe, focusing on the virtues of engineering would lead the engineer to frame and understand the situation in a particular way. It would involve thinking about the ends of engineering and the virtues that would constitute a good engineer. It seems that a virtuous engineer is one who cares about the safety and welfare of society. It also seems that courage is an important virtue for engineers insofar as they have to tell others, such as clients, what they don't want to hear. Clients may not understand technical matters or may give lower priority to safety than they should. So, courage is important for engineers to cultivate in dealing with clients.

Several engineering ethicists have used virtue ethics to think through the virtues specific to engineers. Charles Harris, for example, argues for virtue ethics as a counterbalance to the limitations of codes of ethics.¹ He distinguishes two kinds of virtues in engineering, the technical and the nontechnical. In discussing the

technical excellences of being an engineer, he mentions the obvious importance of mastery of mathematics and physics, engineering science, and design as an important virtue, but he also mentions sensitivity to risk and sensitivity to tight coupling or complex interactions as technical virtues. For the nontechnical virtues, he discusses the importance of techno-social sensitivity, respect for nature, and commitment to public good. According to Harris, these are the dispositions that engineers should cultivate to become good engineers.

Virtue ethics is a third framework for thinking through ethical issues. Ethicists often debate which is the better of the three theories just discussed, but in practice very few people adopt one of these theories whole cloth as the basis for living their lives. The three approaches are better understood as frameworks for ethical analysis. They are different ways of thinking and provide concepts and language that must be interpreted and extended to think through real-world situations.

PRACTICAL ETHICS: ETHICS AS DESIGN

As the autonomous car dilemma suggests, real-world ethical dilemmas are generally more complicated and require much more nuanced and iterative thinking than ethical theories alone can provide. This is both because the details of a situation matter a good deal and because real-world situations take place in time and require multiple and sequential actions. To use a simple example, if you discover something to suggest that your employer is not meeting safety standards, you have to decide whether to report this and to whom. Once you have made that initial decision and acted—say you decide to report it to your supervisor—the problem may not be over. Suppose your supervisor tells you that he already knows and doesn't think it warrants further action. You then must decide whether to do something else or nothing. Even if you do nothing, the situation may change if you find additional evidence that justifies your concern or if others begin to see the problem. The point is that you may have to make decisions and choose actions at many points in an ongoing process.

To understand practical ethical decision-making, many ethicists recognize that although theories are useful, they aren't the be all and end all. Ethical decision-making often requires a form of creativity. Instead of deducing the correct answer from a theory, one must synthesize many elements, make trade-offs, and design a solution.

Caroline Whitbeck developed this conception of practical ethical decision-making, especially in engineering, by making an analogy between practical ethical problems and design problems.² When it comes to design problems, there are solutions that are blatantly wrong but there generally is not a single right answer. That is, there may be several different designs that meet the design criteria and solve the problem. Each solution solves the problem in a different way, and all may be adequate.

Whitbeck asks us to imagine multiple teams of engineers who are tasked with designing a car seat for infants. Imagine that each team is given the same set of specifications for the design—that the car seat should meet regulatory requirements, not weigh more than a certain amount, fit into a certain percentage of automobiles on the road, cost no more than X to manufacture, and so on. Because of the design criteria, certain designs simply will not work—that is, they won't meet the specifications. On the other hand, each design team may come up with a unique design that fits the criteria. The differences in the designs result from different trade-offs. The teams balance cost, weight, size, safety, and ease of manufacturing differently. One team opts for a lighter seat even though it is more expensive (though still within the specs for cost); another team opts for extra safety (beyond what is required) though this will make the seat heavier than it might otherwise be (yet still within the specifications for weight); a third team might opt for ease of use but more difficulty manufacturing; and so on. The point is that although there are some designs that won't work at all, there is no single right or perhaps even best design choice. Each design will have advantages and disadvantages over other designs.

So it is, Whitbeck argues, with ethical decision-making in the real world. In any situation, there are decisions that are clearly wrong. For example, an engineer shouldn't lie about the safety of

a product or take bribes from suppliers. On the other hand, if an engineer discovers an unsafe situation, there is usually more than one possible first step that might be taken to handle the situation. Doing nothing wouldn't be an option, but the engineer might go to her supervisor, or she might go to an ombudsperson or to a senior trusted engineer outside the organization to find out if the senior engineer agrees with the engineer's technical judgment. Depending on the outcome of one of these initial actions, the engineer would have to decide on the next step. Obviously, choices at each step in the process will have advantages and disadvantages that must be balanced, and how they are balanced will depend on such factors as how serious or dangerous the situation is, how time sensitive the problem is, the probability that each course of action would be successful, and so on.

Thinking of practical ethics as design doesn't make ethical decision-making easy. It still requires significant thought and analysis and attention to the details of a situation. Nevertheless, viewing ethical problems as design problems is closer to the real-world experiences of engineers as they try to balance many factors and consider the uncertainties that go with each step that they take. In this way, viewing practical ethical problems on the model of design problems acknowledges that ethical choices are not simple matters of deducing the right answer from a theory. Engineering ethics decision-making involves judgment; solutions and courses of action are often synthetic and creative.

CONCLUSION

So, how should engineers think about ethics? In this chapter we have suggested several ways (not mutually exclusive) that engineers can and should understand the ethical dimensions of their work. First, engineers should think about engineering ethics as professional ethics but not in a narrow sense. Engineering ethics is not just a matter of following professional codes of conduct. At the core, engineering ethics has to do with how engineers behave in the many social relationships involved in engineering and how they consider the social effects and implications of the work that they do. Second,

in dealing with these ethical dimensions, engineers can and should draw on ethical theories. Ethical theories provide a systematic way of thinking about ethics that can be used to understand the situations in which engineers find themselves. They provide frameworks, concepts, and vocabulary that allow engineers to see their circumstances and dilemmas in ways that they might not have comprehended without theoretical analysis. Third, engineers sometimes find themselves in situations in which some options are clearly wrong but the right thing to do is either not obvious or not possible without compromises or trade-offs. So, it is important for engineers to understand that practical ethical decision-making is like design. To figure out what to do in some situations, engineers must design courses of action that involve taking a wide variety of factors into account and acting one step at a time. In the next chapters, we will explore professional values and relationships and how engineers can best balance their many responsibilities.

SUGGESTED FURTHER READING

- Johnson, Deborah G., and Jameson M. Wetmore. "STS and Ethics: Implications for Engineering Ethics." Pages 567–82 in *Handbook of Science and Technology Studies*, 3rd ed. Edited by Edward J. Hackett et al. Cambridge, MA: MIT Press, 2008.
- Rachels, James, and Stuart Rachels. *The Elements of Moral Philosophy*, 8th ed. New York: McGraw-Hill, 2015.
- Whitbeck, Caroline. "Ethics as Design: Doing Justice to Moral Problems." *Hastings Center Report* 26, no. 3 (1996): 9–16.

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